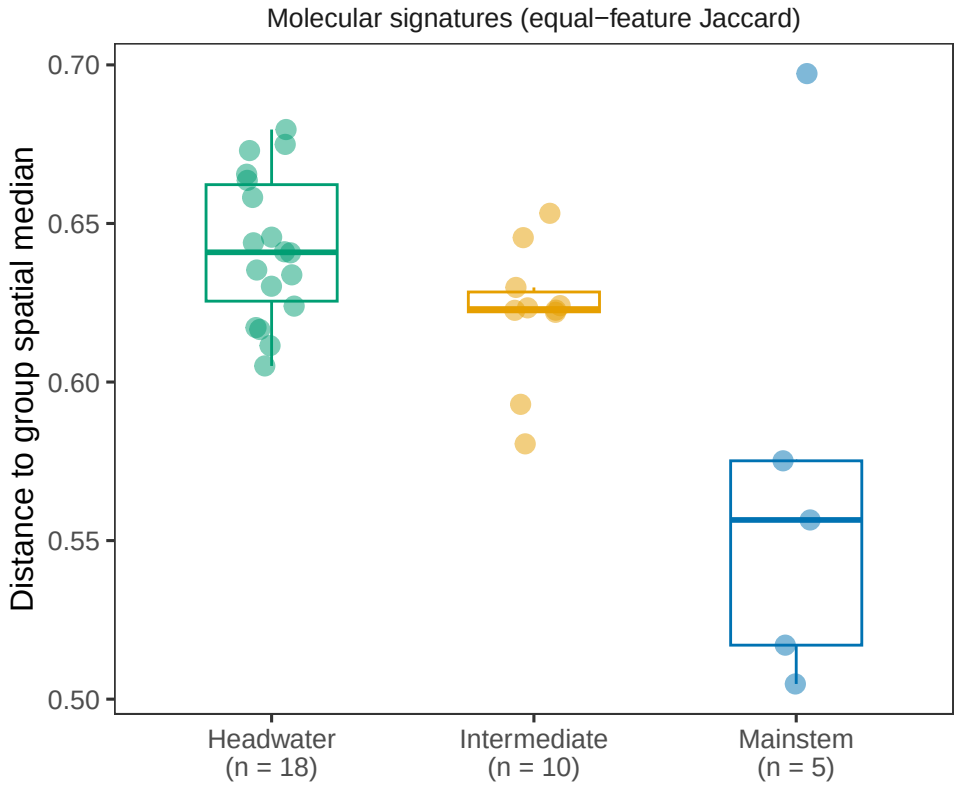
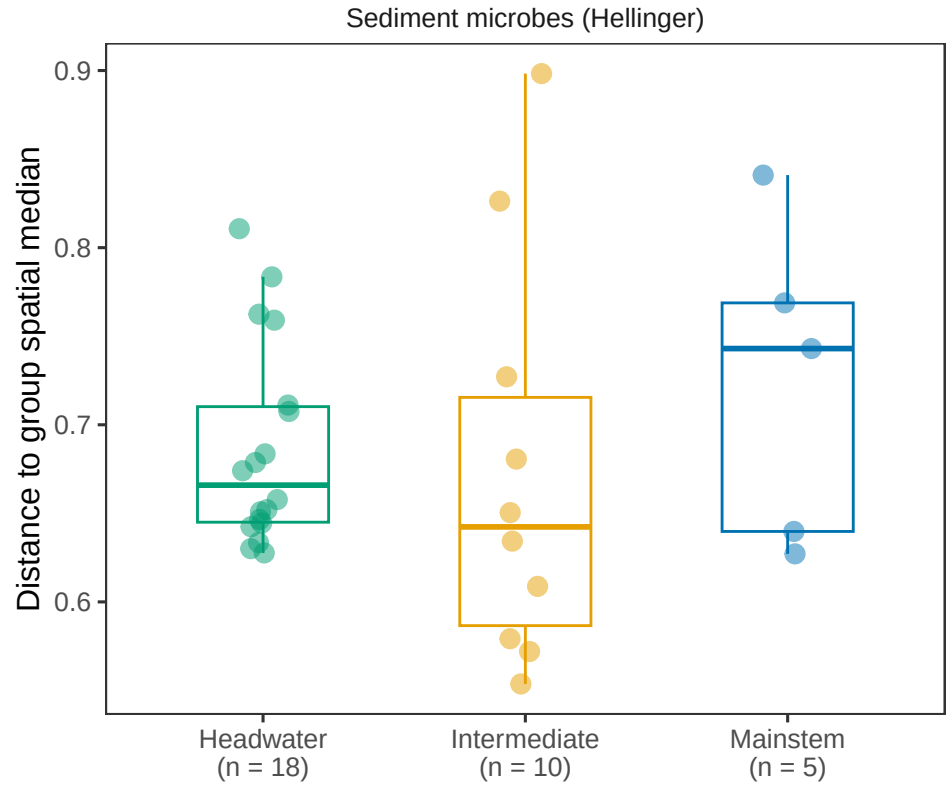
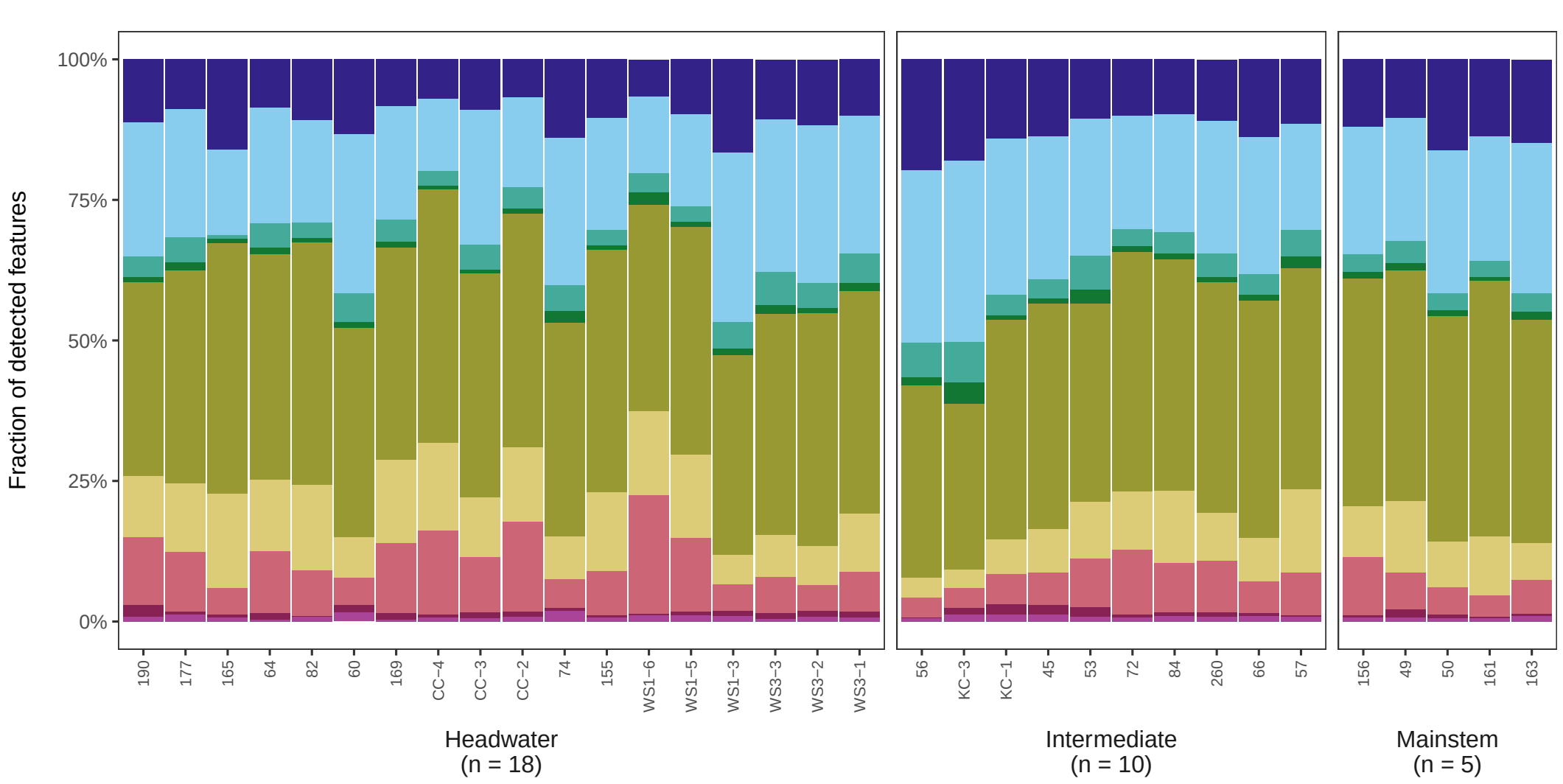
A


Molecular signatures (equal−feature Jaccard)
Distance to group spatial median
0.70
0.65
0.60
0.55
0.50
Headwater
(n = 18)
Intermediate
(n = 10)
Mainstem
(n = 5)

B


Sediment microbes (Hellinger)
Distance to group spatial median
0.9
0.8
0.7
0.6
Headwater
(n = 18)
Intermediate
(n = 10)
Mainstem
(n = 5)

C


Fraction of detected features
100%
75%
50%
25%
0%
190
177
165
64
82
60
169
CC−4
CC−3
CC−2
74
155
WS1−6
WS1−5
WS1−3
WS3−3
WS3−2
WS3−1
56
KC−3
KC−1
45
53
72
84
260
66
57
156
49
50
161
163
Headwater
(n = 18)
Intermediate
(n = 10)
Mainstem
(n = 5)
Lipid−like
Protein−like
Amino−sugar−like
Carbohydrate−like
Lignin−like
Tannin−like
Condensed−hydrocarbon−like
Unsaturated−hydrocarbon−like
Other